

Unit - 1

Basics of Analog & Digital Electronics

- classification of solids on basis of:
 - conductors

They have large no of mobile or free electrons and have high and low resistivity. eg → silver, copper

$$\hookrightarrow \rho = 10^{-6} \Omega\text{-cm}$$

- Insulators

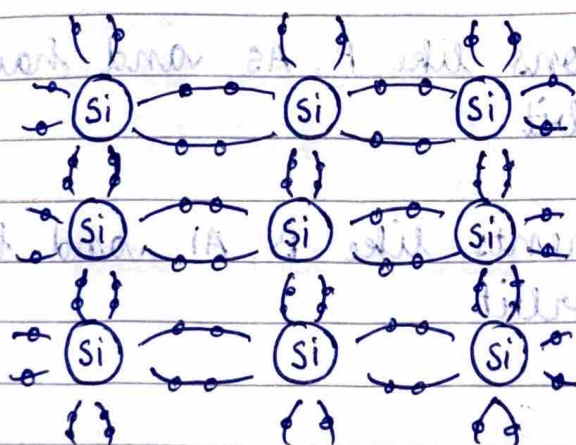
They don't have any free charge carriers or free electrons. They have low conductivity & high resistivity. eg → glass, Rubber, mica

$$\hookrightarrow \rho = 10^{12} \Omega\text{-cm}$$

- Semiconductors

Their conductivity & resistivity lies between conductors and insulators. eg → Germanium

$$\hookrightarrow \rho = 50 \Omega\text{-cm}$$

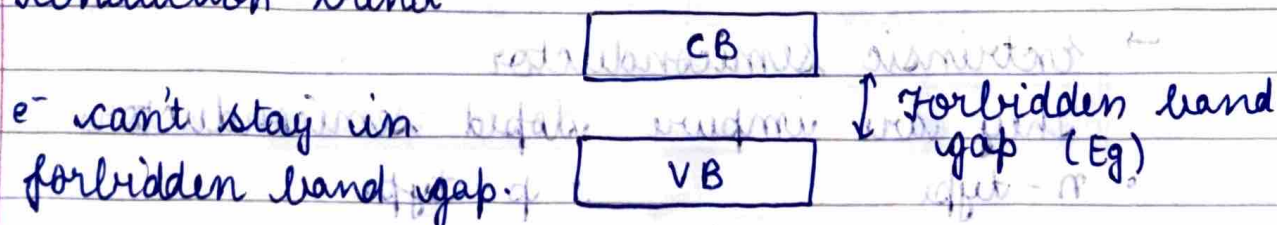


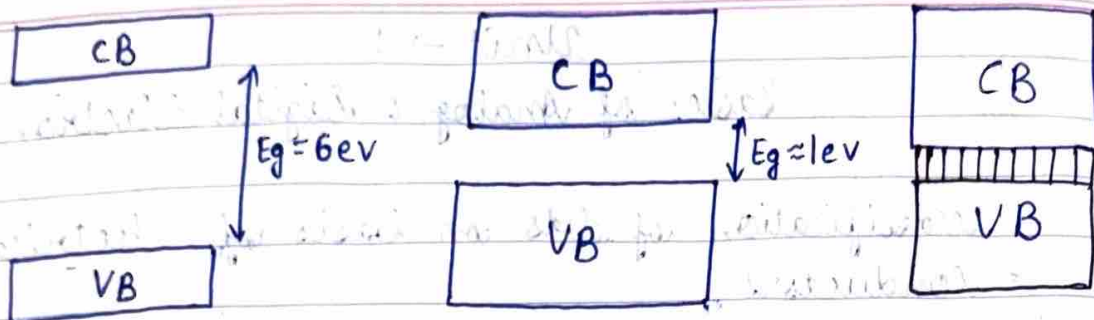
$$\rho_c > \rho_s > \rho_i$$

→ Crystal structure of a silicon atom

- Energy Band diagrams

Due to presence of similar atoms nearby, the valence shell splits into valence band and the conduction band





Insulator

Semiconductor

Conductor

Large band gap

 $E_g \approx 0.75 \text{ eV}$

VB & CB

 $\text{Si} \approx 1.16 \text{ eV}$

overlap

 $\alpha \rightarrow -ve$ $T \uparrow R \downarrow$
 (e^-)

→ Intrinsic semiconductors

They are pure semiconductors and free electrons are only present due to natural causes like thermal energy. $n_e = n_h$ always.

→ Impurities

• Pentavalent

Fifth group elements like P, As and have $5e^-$ in outermost orbit.

• Trivalent

Third group elements like B, Al and have $3e^-$ in outermost orbit.

→ Doping

- Process of adding impurity atoms to pure semiconductors

- In 10 million pure semiconductor atoms, we add 1 atom of impurity.

→ Extrinsic semiconductor

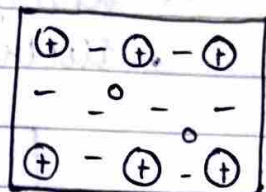
They are impure doped semiconductors.

• n-type

• p-type

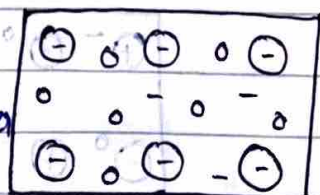
→ N-type semiconductors

- we add pentavalent dopant atom
- electrons act as ^{majority} charge carriers
- dopant atom is electron donor
- $n_e > n_h$



→ P-type semiconductor

- we add trivalent dopant atom
- Holes act as majority charge carriers
- dopant atom is e^- acceptor
- $n_e < n_h$



→ Mass action law

under thermal equilibrium, the products of free electron and free hole concentration is a constant equal to square of intrinsic carrier concentration

no of electrons = no of holes (intrinsic semi)

$$n = p = n_i$$

↳ intrinsic carrier density

By mass action law,

$$\underline{n \cdot p = n_i^2}$$

At room temp, due to thermal energy, there is generation^(G) of e^- -hole pairs but they recombine^(R) very soon to become neutral once again

Also, by thermal equilibrium $G = R$ $\uparrow$ intrinsic

$$\text{also, } R \propto n \text{ and } R \propto p \text{ also } = G = r n_i^2$$

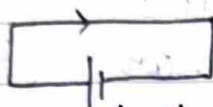
$$R \propto np$$

$$R = (r) \cdot np$$

$$\Rightarrow np = n_i^2$$

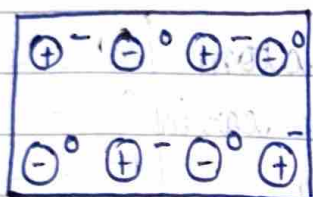
→ Charge densities in a semiconductor

- electrical neutrality



no. of electrons⁺ are equal to
no. of e^- leaving

It's satisfied by conductors & semiconductors
⇒ total +ve charge = total -ve charge



N_D → conc of donor atoms

N_A → conc of acceptor atoms

p → hole conc

n → e^- conc

⇒ For electrical neutrality

$$\underline{N_D + p = N_A + n}$$

Case I Consider n-type material

$$\Rightarrow N_A = 0$$

$$\Rightarrow N_D + p = n$$

$$\Rightarrow N_D \approx n + p$$

But $n \gg p$ for n type material

$$\Rightarrow \underline{N_D \approx n}$$

By mass action law

$$N_D p_n = n_i^2$$

$$\Rightarrow p_n = \frac{n_i^2}{N_D}$$

Case II consider p-type material

$$N_D = 0$$

$$\Rightarrow N_A = p - n$$

$$p \gg n \Rightarrow N_A \approx p$$

By mass action law

$$n_p p_p = n_i^2$$

$$\Rightarrow \underline{n_p = \frac{n_i^2}{N_A}}$$

→ when electric field is applied e^- and holes drift in opposite direction.

V_d is drift velocity for e^-

$$V_d \propto E$$

$$V_d = \mu E$$

μ mobility (constant)

$$\mu = 3800 \text{ cm}^2/\text{V-s} \quad (1.01 \times 88 \cdot (\text{m}^2/\text{V-s}) \times 88131) =$$

$$\text{Also, } I = n e A V_d$$

$$J = \frac{I}{A} = n e V_d = n e \mu E$$

$$\Rightarrow J = \sigma E$$

σ conductivity

also

$$J = (n \mu_n + p \mu_p) e E \Rightarrow \sigma = (n \mu_n + p \mu_p) e$$

Q Mobilities of free e^- and holes in pure germanium are 0.38 and $0.18 \text{ m}^2/\text{V-s}$. Find intrinsic conductivity. $n_i = 2.5 \times 10^{19} / \text{m}^3$

$$\sigma = (n \mu_n + p \mu_p) e$$

$$\sigma_{\text{int}} = (\mu_n + \mu_p) n_i e$$

$$= (0.38 + 0.18) \times 2.5 \times 10^{19} \times 1.6 \times 10^{-19}$$

$$= 0.56 \times 2.5 \times 10^{19} \times 1.6 \times 10^{-19}$$

$$= 2.24 \text{ } (\Omega\text{-m})^{-1}$$

3.7 $\Omega\text{-cm}$ = $3.7 \times 10^{-2} \text{ } (\Omega\text{-m})$ $\rightarrow$ n-type

Q In germanium sample, a donor type impurity is added to extent of 1 atom per 10^8 Ge atoms. Show resistivity drops to $3.7 \text{ } \Omega\text{-cm}$.

$$\mu_n = 3800 \text{ cm}^2/\text{V-s} \quad n_i = 2.5 \times 10^{13} \text{ cm}^3 \quad \mu_p = 1800 \text{ cm}^2/\text{V-s}$$

$$N_{\text{Ge}} = 4.41 \times 10^{22} / \text{cm}^3 \quad e = 1.602 \times 10^{-19} \text{ C}$$

$$\sigma = (n \mu_n + p \mu_p) e$$

$$N_D = \frac{N_{\text{Ge}}}{10^8} = \frac{4.41 \times 10^{22}}{10^8} = 4.41 \times 10^{14} / \text{cm}^3$$

$$\Rightarrow N_D \approx n = 4.41 \times 10^{14}$$

By mass action law

$$P_n = \frac{N_i^2}{N_D} = \frac{6.25 \times 10^{26}}{4.41 \times 10^{14}}$$

$$P_n = 1.41 \times 10^{12}$$

$$\begin{aligned}\sigma &= (3800 \times 4.41 \times 10^{14} + 1800 \times 1.41 \times 10^{12}) \times 1.602 \times 10^{-19} \\ &= (16758 \times 10^{14} + 25.38 \times 10^{14}) \times 1.602 \times 10^{-19} \\ &= 16783.38 \times 1.602 \times 10^{-5} \\ &= 26886.974 \times 10^{-5} \\ &= 0.2688\end{aligned}$$

$$\Rightarrow \rho = 1/\sigma = 1/0.268 = 3.73$$

Q. Consider intrinsic silicon bar of cross section 5 cm^2 and length 0.5 cm at room T. An average field of 20 V/cm is applied at its ends.

$$\mu_n = 1400 \quad \mu_p = 450 \quad n_i = 1.5 \times 10^{10}$$

Find

- e^- & hole components of current density
- Total current
- Resistivity

$$\begin{aligned}a) \quad n &= p = n_i = 1.5 \times 10^{10} \\ E &= 20 \text{ V/cm}\end{aligned}$$

$$\text{hole component} = p \mu_p q E$$

$$J = (n \mu_n + p \mu_p) q E$$

$$\text{electron component} = n \mu_n q E$$

$$\text{elec comp} = 1.5 \times 10^{10} \times 1400 \times 1.6 \times 10^{-19} \times 20$$

$$= 67.20 \text{ } \mu\text{A/cm}^2$$

$$\text{Hole comp} = 1.5 \times 10^{10} \times 450 \times 1.6 \times 10^{-19} \times 20$$

$$= 21.6 \text{ } \mu\text{A/cm}^2$$

$$b) \Rightarrow J = 67.20 + 21.60$$

$$J = 78.80 \text{ } \mu\text{A/cm}^2$$

$$\text{Now } I = JA$$

$$I = 5 \times 78.8$$

$$\Rightarrow I = 394.0 \mu\text{A}$$

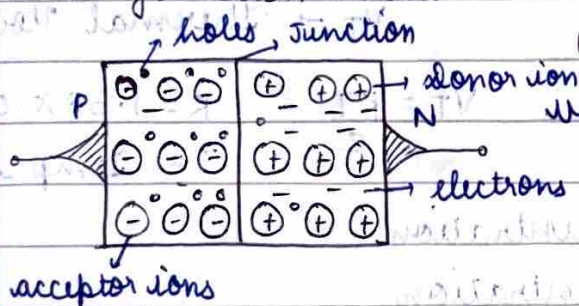
$$c) \text{ Now, } J = \sigma E$$

$$10^{-6} \times 78.8 = \sigma \times 20$$

$$\sigma = 3.94 \times 10^{-6}$$

$$\Rightarrow \rho = \frac{1}{\sigma} = 22.52 \times 10^4$$

→ P-N junction diode



Biassing → Application of external voltage across two terminals

1. No Bias

The concentration of holes on p side is greater than on n side so holes diffuse from p side to n side and forms a diffusion current

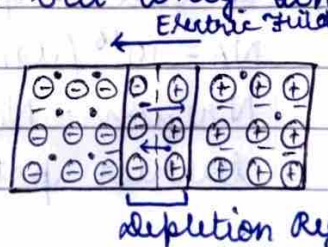
Similarly electrons from N side move towards the junction. The holes from P & e^- from N side recombine and immobile ions surface out

→ Depletion Region

Region having no mobile charges but only uncovered immobile ions are present.

→ Barrier potential

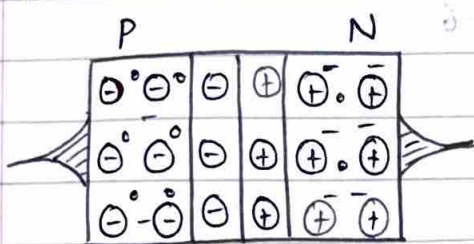
stops further diffusion into depletion region



- Drift current (opposite dir. of $I_{\text{diffusion}}$)
Produced due to drifting of minority charge carriers in depletion layer
under steady state

Due to $\leftarrow I_{\text{drift}} = I_{\text{diffusion}} \rightarrow$ due to majority charge carriers
minority net current = 0
charge carriers

- Barrier Potential (V_b)



$$V_b = \frac{KT}{e} \ln \left(\frac{N_A N_D}{N_i^2} \right)$$

$V_T \rightarrow$ Thermal Voltage

$$V_T = \frac{KT}{e}$$

$$K = 1.38 \times 10^{-23} \text{ J/K}$$

$T =$ Temp in K

$N_A \rightarrow$ acceptor concentration

$N_D \rightarrow$ donor concentration

$N_i \rightarrow$ con^c of intrinsic charge carriers

At room temp, $T = 27^\circ\text{C} = 300\text{K}$

$$V_T = \frac{KT}{e} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 0.026 \text{ V}$$

$$V_T = 0.026$$

$\rightarrow$ at room temp

$$\Rightarrow V_b = 0.026 \ln \left(\frac{N_A N_D}{N_i^2} \right)$$

Q. A silicon p-n junction at room temp doped at $N_A = 10^{16} / \text{cm}^3$ in p region & $N_D = 10^{17} / \text{cm}^3$ in n region. $N_i = 1.5 \times 10^{10} / \text{cm}^3$. Calculate built in barrier potential.

$$V_B = 0.026 \times \ln \left(\frac{10^{16} \times 10^{17}}{(1.5 \times 10^{19})^2} \right)$$

$$V_B = 0.026 \ln \left(\frac{4 \times 10^{-3}}{9} \right)$$

$$\underline{V_B = 0.757} \quad (\text{silicon at room temp})$$

→ width of depletion region (W_D)

$$W_D = \sqrt{\frac{2E}{q} \left[\frac{1}{N_A} + \frac{1}{N_D} \right] V_B}$$

$\epsilon \rightarrow$ electrical permittivity

$$\epsilon = \epsilon_r \epsilon_0$$

For Si, $\epsilon = 1.04 \times 10^{-12} \text{ F/cm}$

For Ge, $\epsilon = 1.2448 \times 10^{-12} \text{ F/cm}$

Q. A silicon p-n junction at room temp. $N_A = 10^{16} / \text{cm}^3$

$N_D = 10^{17} / \text{cm}^3$, $N_A = 1.5 \times 10^{19} / \text{cm}^3$. $W_D = ?$

Here $V_B = 0.757$

$$W_D = \sqrt{\frac{2 \times 1.04 \times 10^{-12}}{1.6 \times 10^{-19}} \left[\frac{1}{10^{16}} + \frac{1}{10^{17}} \right] 0.757}$$

$$= \sqrt{1.3 \times 10^7 (10^{-17} (11)) 0.757}$$

$$= \sqrt{0.98411 \times 10^{-10}}$$

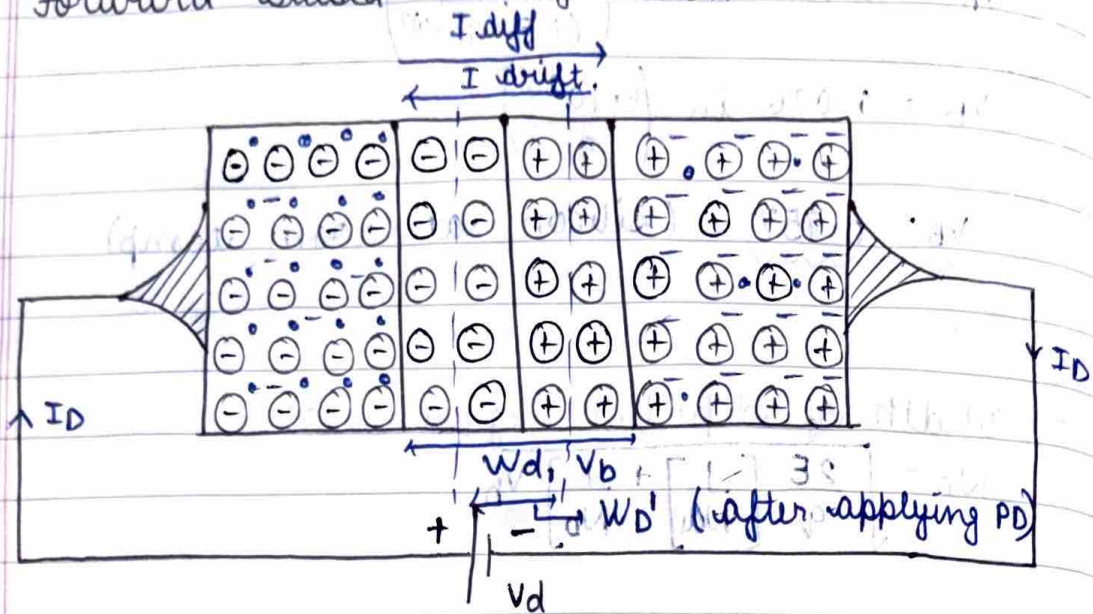
$$= \sqrt{10.825 \times 10^{-10}}$$

$$= 3.290 \times 10^{-5} \text{ cm}$$

$$= 0.329 \mu\text{m}$$

↑ reverse bias potential ↑ T

Forward Biased P-N junction



On applying positive terminal of battery on p side, the +ve terminal repels the holes in p side and they enter into depletion region. Similarly negative terminal pushes electrons from N side towards depletion region. These holes and electrons recombine with immobile ions. Hence the width of depletion layer decreases.

$$V_D = 0$$

$$W_D$$

$$V_D > 0$$

$$W_D'$$

$$W_D' < W_D$$

$$\underline{V_D \uparrow \quad W_D \downarrow}$$

$$I_D = -I_{\text{drift}} + I_{\text{diff}}$$

$$\underline{I_D = I_{\text{diffusion}} - I_{\text{drift}}}$$

$T \uparrow$ minority chg carrier $\uparrow$

When we apply forward potential

$$W_D \downarrow \quad V_b \downarrow$$

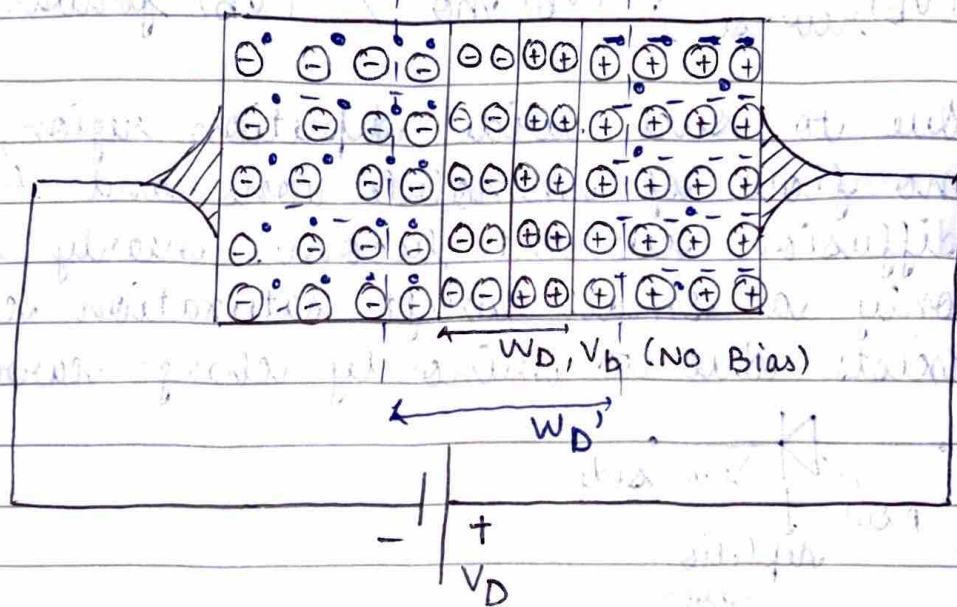
$$\underline{V_b' = V_b - V_D}$$

For Si, $V_b = 0.7$ when $V_D = 0.7$

$$\Rightarrow V_b' = 0$$

Since $V_b' = 0$, a flood of electrons pass through junction and an exponential growth is seen in forward current

→ Reverse Bias condition



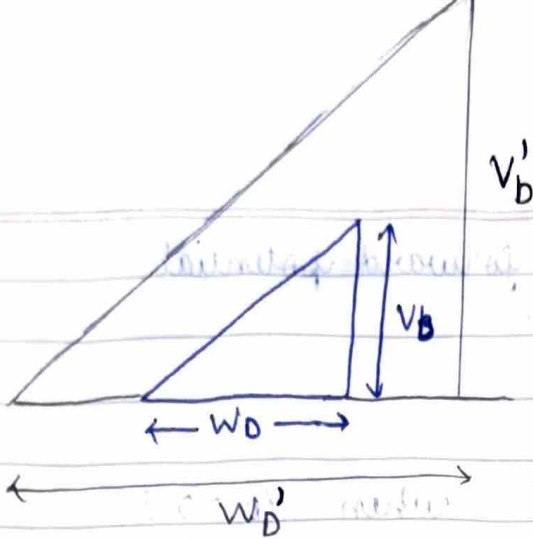
On applying negative terminal of battery on p side the -ve terminal attracts the holes from p side & similarly positive terminal attracts electrons from n side and there is uncovering of immobile ions and leads to widening of width of depletion region.

$$V_D = 0$$

$$W_D$$

$$V_D > 0$$

$$W_D'$$



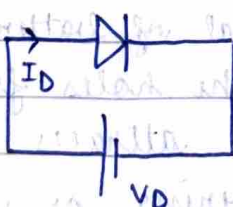
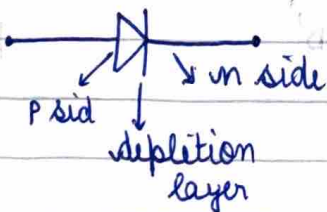
$$V_D \uparrow W_D \uparrow V_b \uparrow$$

$$\underline{V_D' = V_b + V_D}$$

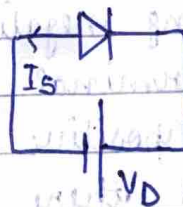
$$(W_D)_{\text{reverse}} > (W_D)_{\text{no bias}} > (W_D)_{\text{forward bias}}$$

$$(V_b)_{\text{reverse}} > (V_b)_{\text{no}} > (V_b)_{\text{forward}}$$

Due to such wide depletion region, there is no flow of immobile ions and hence the diffusion current becomes nearly zero. So only a small drift saturation current exists due to minority charge carriers.



Forward



Reverse

$I_D \rightarrow$ Forward current

$I_S \rightarrow$ Reverse saturation current

→ Diode Current-Voltage Relation

$$I_D = I_S \left(e^{\frac{KV_D}{TK}} - 1 \right)$$

↑ Boltzmann constant

$$K = 11600$$

$\eta \rightarrow$ ideality factor

$\eta = 1$ for Ge & Si
at high I_D

$\eta = 1$ (Ge) } low I_D
 $\eta = 2$ (Si) }

Also, $\frac{K}{TK} = \frac{1}{V_T \eta}$

$$V_T = \frac{TK}{11600}$$

$$\Rightarrow I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

Q A germanium diode displays a forward voltage of 0.25 V at 10 mA current at room temp. Find reverse saturation current.

$I_D = 10 \text{ mA}$, $V_D = 0.25 \text{ V}$, $T_C = 27^\circ \text{C}$, $TK = 300 \text{ K}$

$$V_T = \frac{TK}{11600} = \frac{300}{11600} = 0.026 \text{ V}, \quad \eta = 1$$

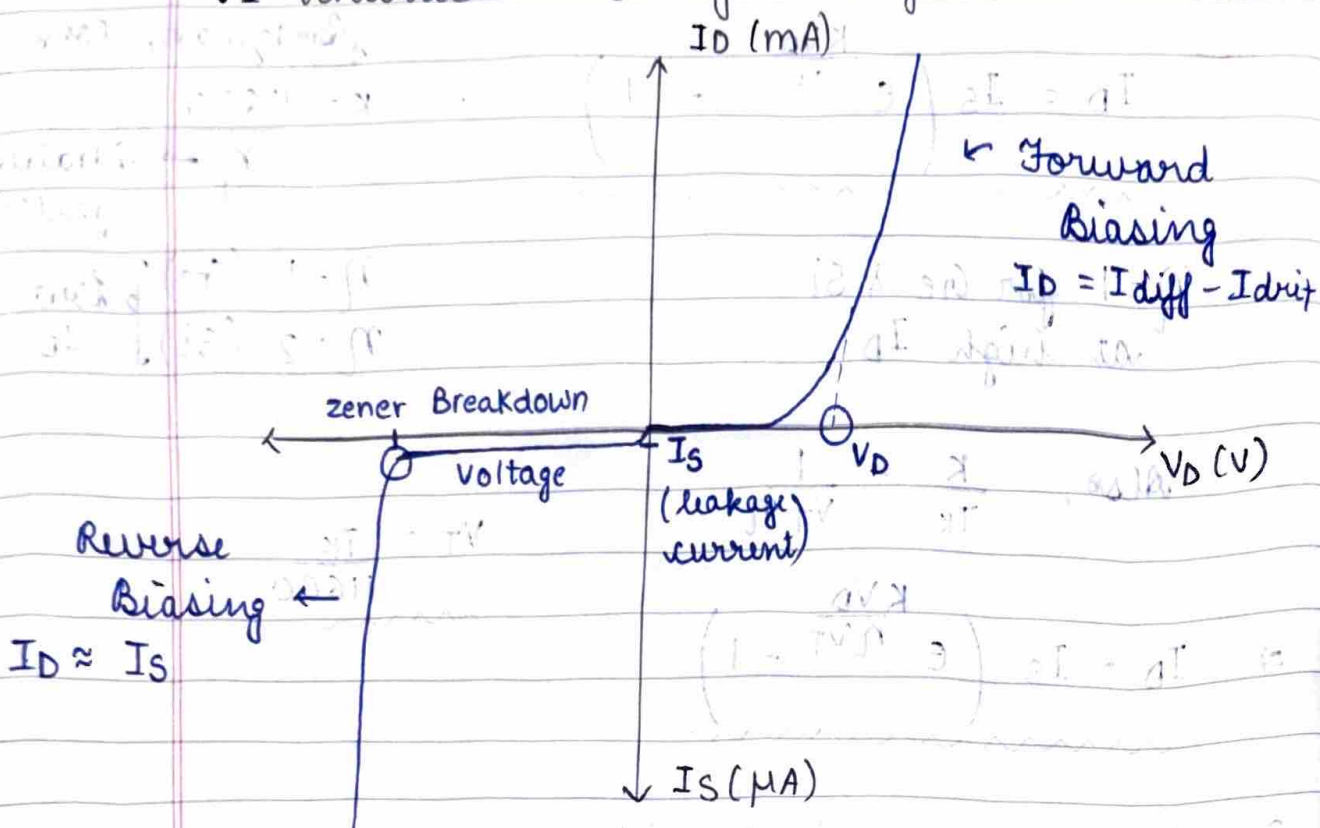
$$\Rightarrow 10 = I_S \left(e^{\frac{0.25}{0.026}} - 1 \right)$$

$$\Rightarrow 10 = I_S (e^{9.615} - 1) = I_S (14987.922 - 1)$$

$$\Rightarrow I_S = \frac{10}{14986.922} = 6.67 \times 10^{-4} \text{ mA}$$

$$\Rightarrow I_S = 0.667 \mu\text{A}$$

VI characteristics of P-N junction diode



On increasing reverse bias voltage to high value, electrons will acquire a high kinetic energy & they will break the covalent bond and become free. These free electrons collide with others and make them free earlier in a chain reaction. This leads to breakdown of the diode.

→ Peak inverse voltage

Maximum reverse voltage that can be applied to diode before it breaks down

→ Effects of Temp on V-I characteristics

• Forward bias

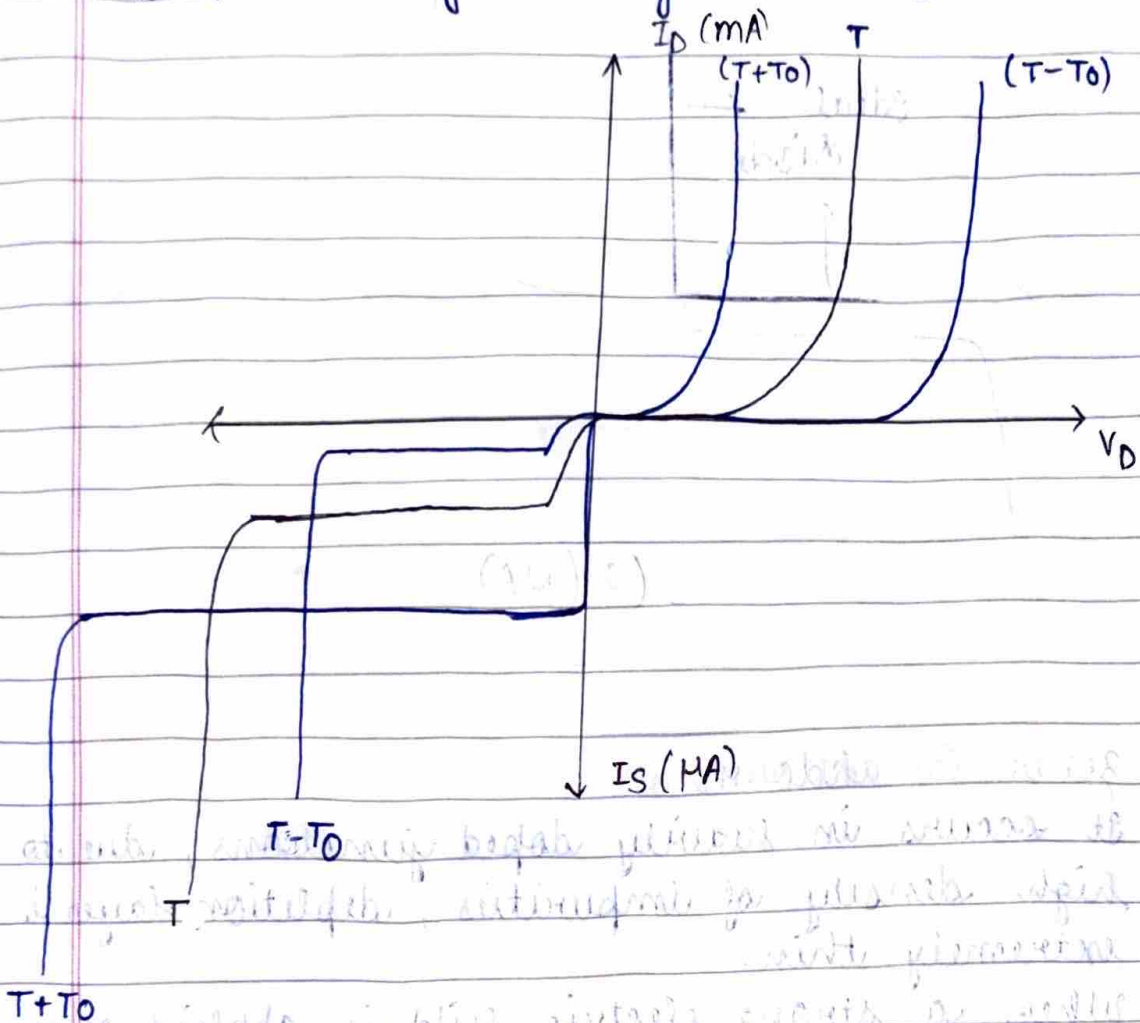
Graph shifts left at rate of $2.5 \text{ mV}/^\circ\text{C}$ rise in temperature

$$\text{Temp} \uparrow \quad V_D \downarrow$$

$$\text{Temp} \downarrow \quad V_D \uparrow$$

Reverse Bias

I_s doubles for every 10°C rise in temp



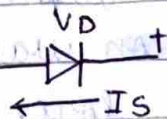
→ ideal v/s practical diode



$$I_D = I_{\text{diff}} - I_{\text{drift}}$$

$$\Rightarrow I_D = I_{\text{maj}} - I_s$$

$V_D \rightarrow$ very small



$$I_D \approx I_s$$

$V_D \rightarrow$ very large

conditions of practical diode

(R.B)

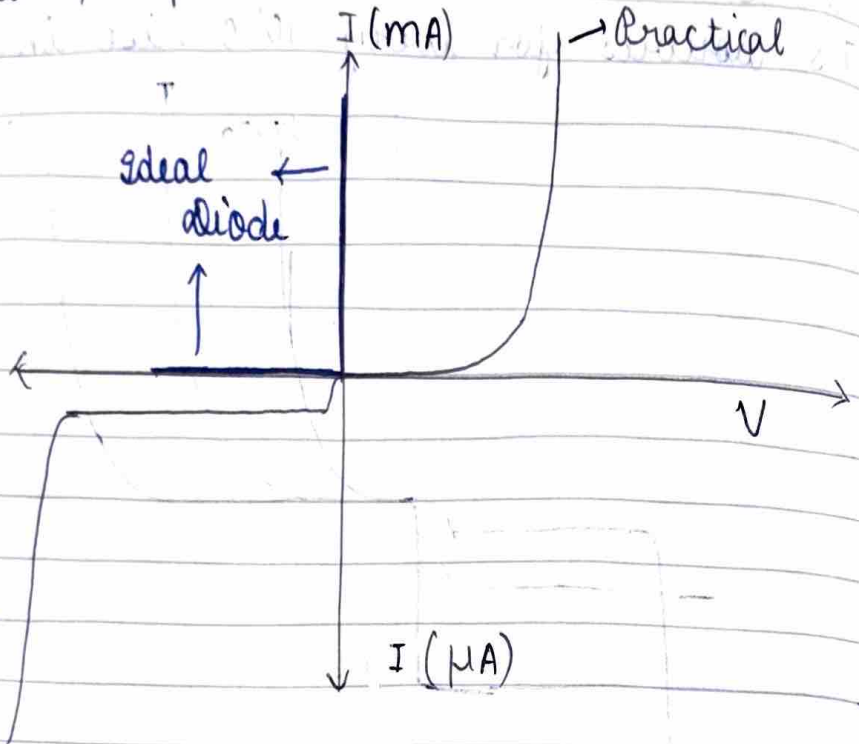
$$\mu_D = 0 \text{ (F.B)}$$

resistance of diode

$$I_s \approx 0, \mu_D = \infty$$

conditions of ideal diode

→ Ideal V/s practical



→ Zener Breakdown

- It occurs in heavily doped junctions, due to high density of impurities, depletion layer is extremely thin.
- When a strong electric field is applied across this thin depletion layer, it exerts a force directly on valence electrons of host atoms and this force pulls electrons out of their covalent bonds creating large no. of free charge carriers.
- It occurs at low reverse voltages ($< 5V$) and has negative temp coefficient. ($T \uparrow V_b \downarrow$)

→ Avalanche Breakdown

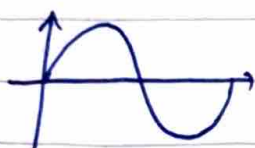
- It occurs in lightly doped junctions having a wider depletion region.
- This is collision based process. Free electrons in depletion region are accelerated by electric field.

Because depletion layer is wide, they gain enough kinetic energy to knock other electrons (valence) out of their bonds during collision

- It occurs at high reverse voltages ($> 8V$) and have a positive temperature coefficient ($T \uparrow V_B \uparrow$)

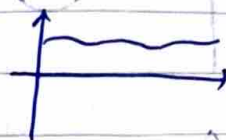
→ Rectification

conversion of AC into pulsating DC is rectification



AC

Rectifier →



DC

To get straight DC we use filters

AC → Transformer → Rectifier → Filter → Voltage Regulator → Load → DC

→ Rectifier

1. Half wave
2. Full wave
 - a) Centre tapped
 - b) Bridged

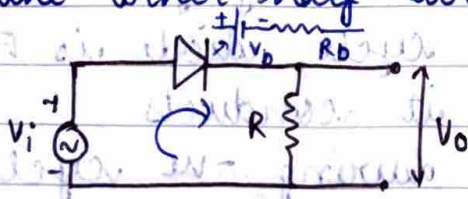
Disadvantage of HW

- Only one half cycle is conducting

• $\gamma = 121\%$ and $\eta = 40.5\%$

→ Half wave Rectifier

only one half of the AC voltage is rectified, for the other half we get zero voltage



Diode is forward biased

Applying KVL

$$V_i - V_b - I R_D - I R = 0$$

$$I = \frac{V_i - V_b}{R_D + R}$$

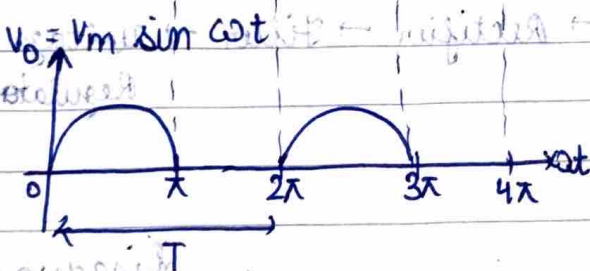
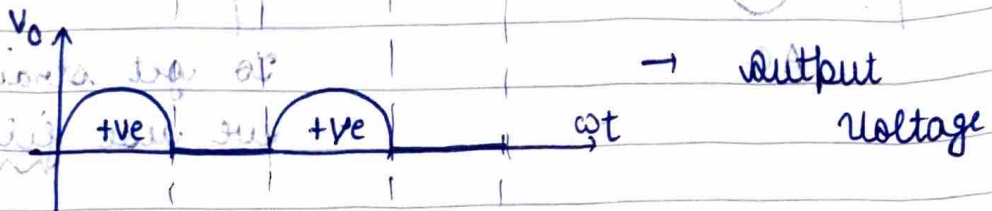
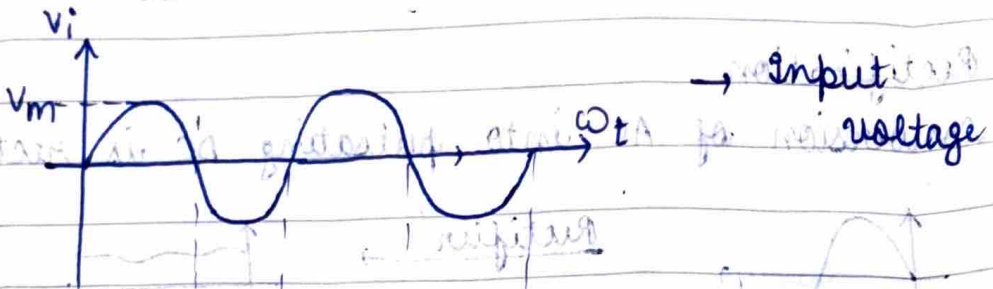
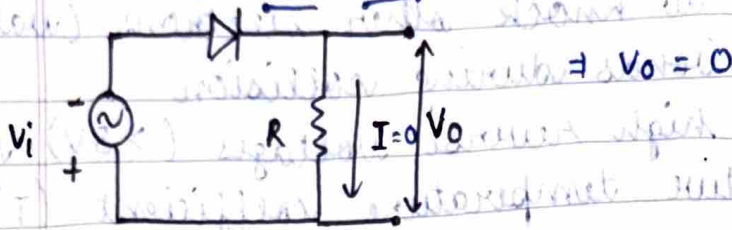
$$V_o = I R$$

$$V_o = \left(\frac{V_i - V_b}{R_D + R} \right) R$$

$$R_D \ll R$$

$$\Rightarrow V_o = V_i - V_b$$

During -ve half cycle, diode is reverse biased



$$V_{avg} = \left(\int_0^{2\pi} V_o d(\omega t) \right) \div (2\pi - 0)$$

$$= \left(\int_0^{2\pi} V_m \sin \omega t d(\omega t) \right) \div 2\pi$$

$$= \left(\frac{V_m}{2\pi} \int_0^{2\pi} \sin A dA \right)$$

$$= -\frac{V_m}{2\pi} \cos \omega t \Big|_0^{2\pi}$$

$$= -\frac{V_m}{2\pi} (1 - 1) = 0$$

During first half cycle, diode is F.B it conducts
During -ve cycle diode is R.B, is replaced by open circuit and doesn't conduct

1. Average voltage

$$\Rightarrow V_{avg} = \frac{1}{2\pi} \int_0^{2\pi} v_o d(\omega t)$$

$$= \frac{1}{2\pi} \int_0^{\pi} V_m \sin \omega t d(\omega t)$$

$V_o = 0$ for $\pi - 2\pi$

$$\Rightarrow V_{avg} = \frac{V_m}{2\pi} (-\cos \omega t) \Big|_0^{\pi} = \frac{V_m}{2\pi} (1 - (-1)) = \frac{2V_m}{2\pi}$$

$$\Rightarrow V_{avg} = \frac{V_m}{\pi} = 0.318 V_m$$

2. Average current

$$I_{avg} = \frac{V_{avg}}{R} = \frac{V_m}{\pi R} = \frac{I_m}{\pi} = I_{avg} = 0.318 I_m$$

3. RMS current

$$\Rightarrow I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I^2 d(\omega t)}$$

$I_o = 0$ $\pi \leq \omega t \leq 2\pi$

$$= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} 1 - \cos 2\omega t d(\omega t)} = \sqrt{\frac{I_m^2}{4\pi} (\omega t - \sin 2\omega t) \Big|_0^{\pi}}$$

$$= \sqrt{\frac{I_m^2}{4\pi} (\pi - 0)} = \frac{I_m}{2}$$

$$\Rightarrow I_{rms} = \frac{I_m}{2}$$

4. RMS voltage

$$V_{rms} = I_{rms} \times R = \frac{I_m}{2} \times R = \frac{V_m}{2}$$

$$\Rightarrow V_{rms} = \frac{V_m}{2}$$

- Significance of RMS value
A bulb will shine same when either 5V DC or 5V RMS AC is applied to it

$$\text{RMS value} \leq \text{Peak value}$$

$$I_{\text{RMS}} \leq I_m$$

$$V_{\text{RMS}} \leq V_m$$

RMS is the DC equivalent of AC circuits

5. Form Factor → Tells about output wave form
It's defined as ratio of RMS voltage and average load voltage

$$F.F = \frac{V_{\text{RMS}}}{V_{\text{avg}}}$$

$$FF \geq 1$$

$$FF = \frac{V_m}{2} \times \frac{\pi}{2} = \frac{\pi}{2}$$

$$FF = \frac{\pi}{2}$$

$$FF = 1.575$$

6. Ripple Factor (γ)
Output current contains both AC & DC comp.
Ripple factor measures % of AC component in rectified output

$$\gamma_{\text{ideal}} = 0\%$$

$$\gamma = \frac{\text{RMS value of AC component}}{\text{average output value}}$$

$$I_o = I_{\text{AC}} + I_{\text{DC}}$$

$$I_{\text{AC}} = I_o - I_{\text{DC}}$$

$$\Rightarrow I_{AC\text{ RMS}} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_{AC}^2 d(\omega t)} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_0^2 + I_{DC}^2 - 2I_0 I_{DC} d(\omega t)}$$

$$= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} I_0^2 d(\omega t) + \frac{1}{2\pi} \int_0^{2\pi} I_{DC}^2 d(\omega t) - \frac{1}{\pi} \int_0^{2\pi} I I_{DC} d(\omega t)}$$

$$= \sqrt{I_{0\text{ RMS}}^2 + I_{DC}^2 - 2I_{DC}^2} = \sqrt{I_{0\text{ RMS}}^2 - I_{DC}^2}$$

$$\Rightarrow \gamma = \frac{\sqrt{I_{0\text{ RMS}}^2 - I_{DC}^2}}{I_{DC}} = \sqrt{\frac{I_{0\text{ RMS}}^2}{I_{DC}^2} - 1}$$

$$\Rightarrow \gamma = \sqrt{(FF)^2 - 1}$$

$$\Rightarrow \gamma = 1.21$$

7. Efficiency (η)

It's defined as ratio of DC power available at load to input AC power

$$\eta \% = \frac{P_{\text{load}}}{P_{\text{in}}} \times 100$$

$$\Rightarrow \eta = \frac{I_{DC}^2}{I_{\text{RMS}}^2} \times 100 = \frac{I_m^2}{\pi^2} \times \frac{4}{I_m^2} \times 100$$

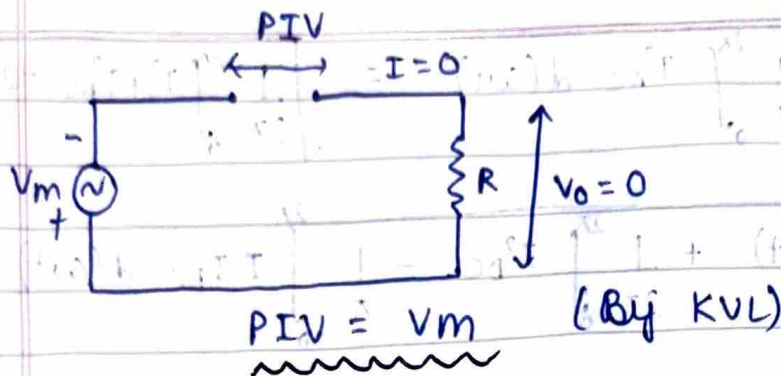
$$\Rightarrow \eta = \frac{4}{\pi^2} \times 100\%$$

$$\Rightarrow \eta = 40.56\%$$

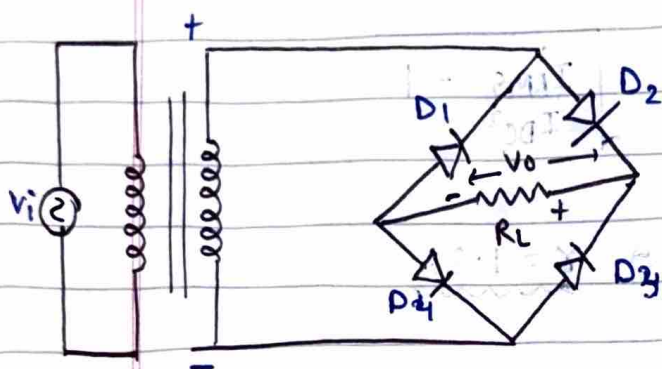
8. Peak inverse voltage (PIV)

$$PIV \geq V_m$$

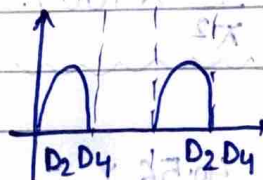
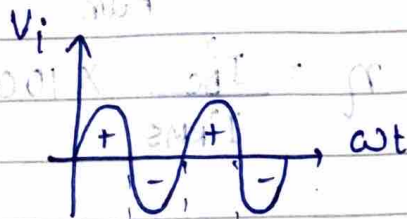
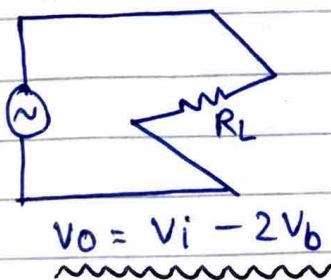
It is the maximum reverse bias voltage that can be applied across the diode before entering into Zener/breakdown region



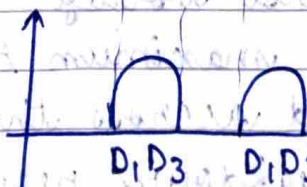
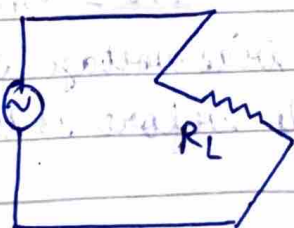
→ Full wave Bridge Rectifiers

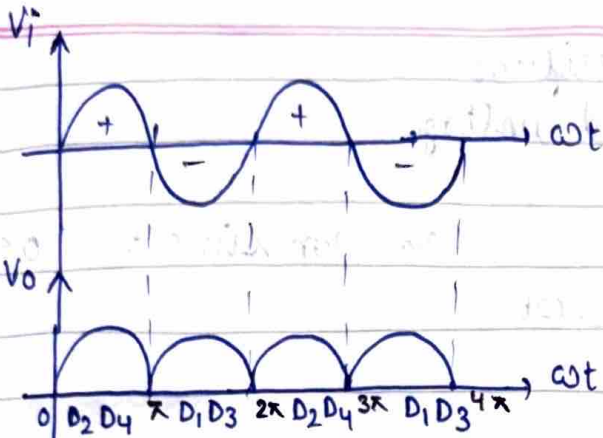


During first positive half cycle, diode D_2 and D_4 are forward biased and conduct while D_1 & D_3 are RB and replaced by open circuit.

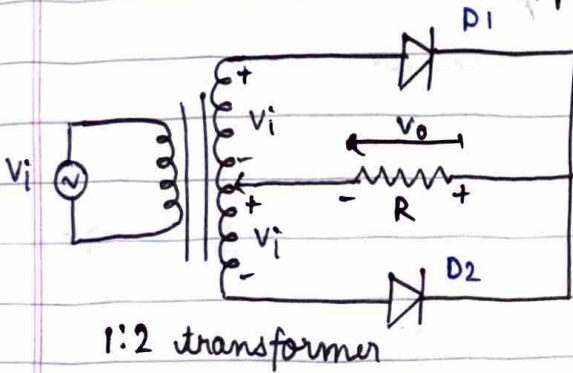


During negative half cycle $D_1, D_3 \rightarrow FB$ and $D_2, D_4 \rightarrow RB$

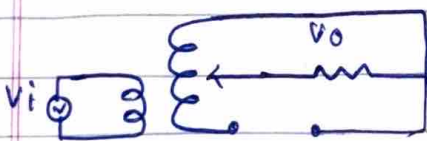




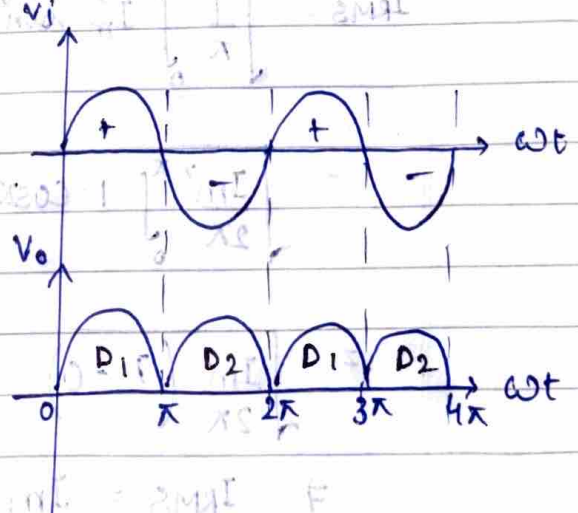
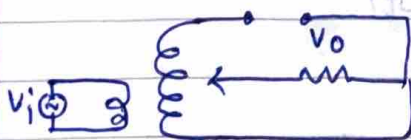
→ Full wave centre tapped rectifiers



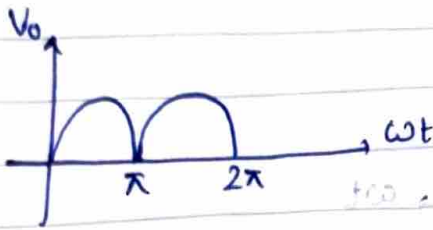
during +ve half cycle $D_1 \rightarrow FB$ & $D_2 \rightarrow RB$



during -ve half cycle $D_2 \rightarrow FB$ & $D_1 \rightarrow RB$



- Full wave Rectifiers
 → Average (D.C) load voltage



$$V_o = V_m \sin \omega t$$

$$0 \leq \omega t \leq \pi$$

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t = -\frac{V_m}{\pi} \cos \omega t \Big|_0^{\pi}$$

$$= -\frac{V_m}{\pi} [-1 - 1] = \frac{2V_m}{\pi}$$

$$\Rightarrow \underline{V_{avg} = \frac{2V_m}{\pi}}$$

- Avg load current

$$V_{avg} = I_{avg} \times R$$

$$I_{avg} = \frac{2V_m}{\pi R} = \frac{2I_m}{\pi}$$

$$\Rightarrow \underline{I_{avg} = \frac{2I_m}{\pi}}$$

- RMS load current

$$I_{RMS} = \sqrt{\frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \omega t d(\omega t)}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} 1 - \cos 2\omega t} = \sqrt{\frac{I_m^2}{2\pi} \left(\omega t - \frac{\sin 2\omega t}{2} \right) \Big|_0^{\pi}}$$

$$= \sqrt{\frac{I_m^2}{2\pi} (\pi - 0)} = \frac{I_m}{\sqrt{2}}$$

$$\Rightarrow \underline{I_{RMS} = \frac{I_m}{\sqrt{2}}}$$

→ RMS voltage

$$V_{RMS} = I_{RMS} \times R$$

$$V_{RMS} = \frac{I_m \times R}{\sqrt{2}}$$

$$\Rightarrow V_{RMS} = \frac{V_m}{\sqrt{2}}$$

→ Form Factor

$$FF = \frac{V_{RMS}}{V_{avg}} = \frac{V_m}{\sqrt{2}} \times \frac{\pi}{2V_m}$$

$$FF = \frac{\pi}{2\sqrt{2}} \quad FF = 1.11$$

→ Ripple Factor

$$\gamma = \sqrt{(FF)^2 - 1} \times 100\%$$

$$\gamma = \sqrt{(1.11)^2 - 1} \times 100$$

$$\gamma = 48.1\%$$

$$\gamma_{FWR} < \gamma_{HWR}$$

FWR is more favourable

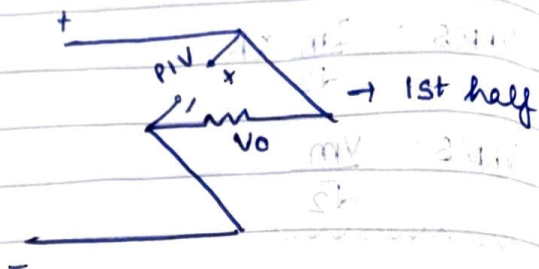
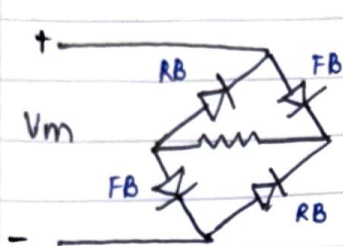
→ Efficiency

$$\eta \% = \frac{P_{out}}{P_{in}} \times 100$$

$$= \frac{I_{DC}^2 R}{I_{RMS}^2 R} \times 100 = \frac{4}{\pi^2} \times 100 = \frac{8}{\pi^2} \times 100\%$$

$$\eta = \frac{8}{\pi^2} \times 100\%$$

→ PIV for full wave bridge rectifier



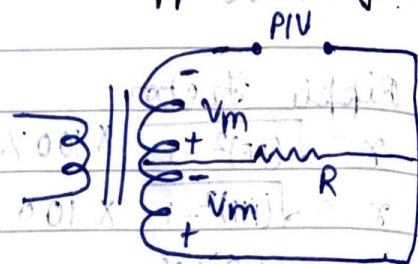
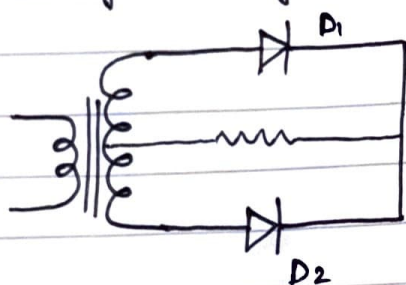
$$V_o = V_m$$

$$-V_o + \text{PIV} = 0$$

$$\text{PIV} = V_m$$

$$\text{PIV} \geq V_m$$

→ PIV for ~~half~~ full wave centre tapped rectifier



$$+V_m - V_o = 0$$

$$V_m = V_o$$

$$V_o - \text{PIV} + V_m = 0$$

$$\text{PIV} = 2V_m$$

$$\text{PIV} \geq 2V_m$$

Property

HWR

FWCTR

FWBR

V_{avg}

$$V_m/\pi$$

$$2V_m/\pi$$

$$2V_m/\pi$$

I_{avg}

$$I_m/\pi$$

$$2I_m/\pi$$

$$2I_m/\pi$$

V_{RMS}

$$V_m/2$$

$$V_m/\sqrt{2}$$

$$V_m/\sqrt{2}$$

I_{RMS}

$$I_m/2$$

$$I_m/\sqrt{2}$$

$$I_m/\sqrt{2}$$

F.F.

$$1.575 (\pi/2)$$

$$1.11 (\pi/2\sqrt{2})$$

$$1.11$$

η

$$40.56\%$$

$$81.13\%$$

$$81.13\%$$

γ

$$1.21$$

$$0.481$$

$$0.481$$

PIV

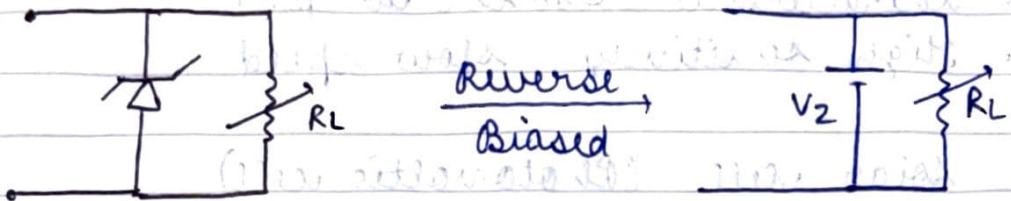
$$V_m$$

$$2V_m$$

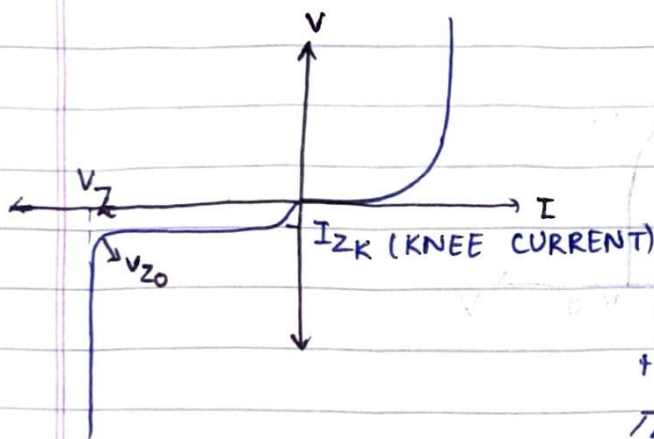
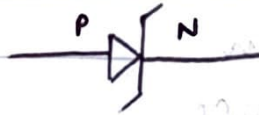
$$V_m$$

→ Zener diode

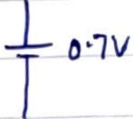
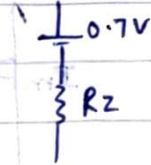
- It's special diode widely used for voltage regulation across varying load



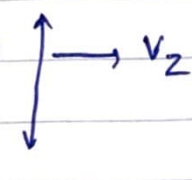
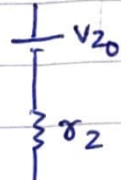
- They have high power dissipation capabilities in reverse biasing



F.B



R.B



→ Light emitting Diodes (LED)

- Emits light when forward biased
- Convert electrical energy directly to light energy
- VI curve is similar to standard diode

→ Liquid Crystal Display (LCD)

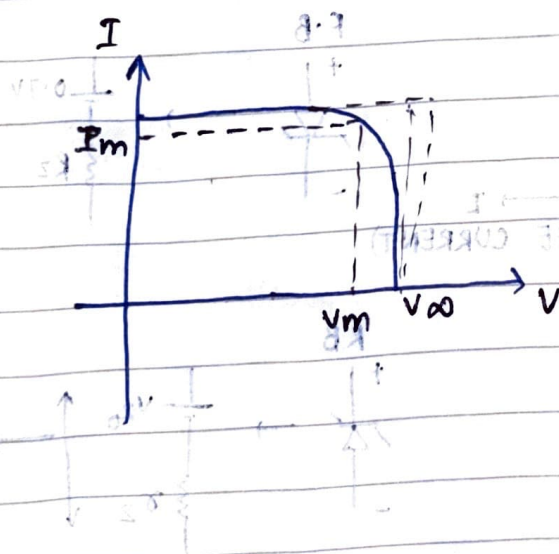
- Uses back light to illuminate pixels rather than emitting light directly
- Consumes more power

→ Photo transistor

- Converts light energy to electrical energy
- It's type of BJT which uses light rather I_B
- Characteristics similar to BJT
- High sensitivity, slow speed

→ Solar cell (Photovoltaic cell)

- Converts light to electrical energy using photovoltaic effect
- Unique V-I characteristics
- Renewable & clean
- Low maintenance



(131) which generates light

by using solar energy.

It is a type of solar cell which converts solar energy into electrical energy.

It is a type of solar cell which converts solar energy into electrical energy.

(132) which generates light

by using solar energy.

It is a type of solar cell which converts solar energy into electrical energy.

It is a type of solar cell which converts solar energy into electrical energy.